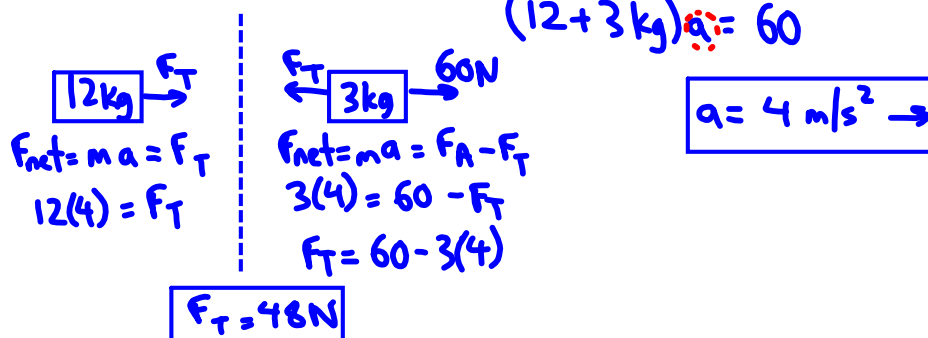
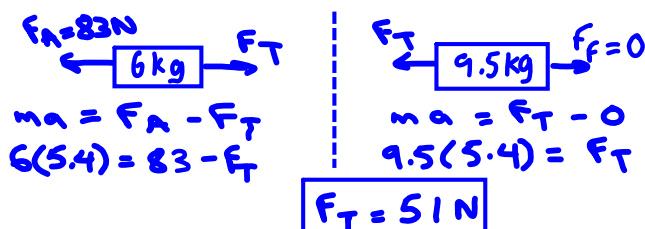


Forces II Practice

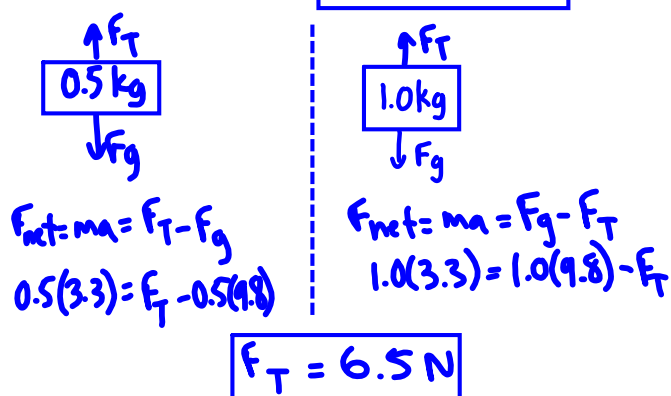
1. a) $u=0 \Rightarrow F_F=0 \Rightarrow F_{\text{net}} = (m_1 + m_2)a = F_A - 0$
 $(12 + 3 \text{ kg})a = 60$



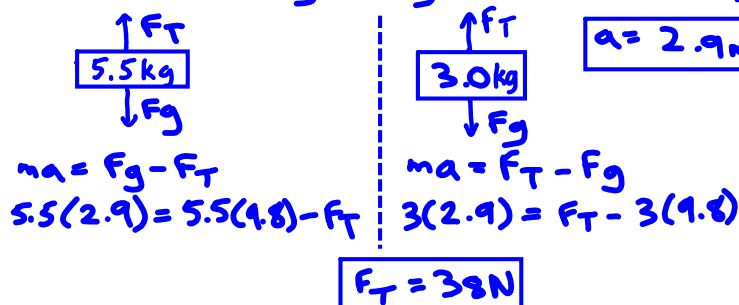
b) $F_{\text{net}} = ma = F_A - 0 \Rightarrow (6.0 + 9.5)a = 83 \Rightarrow a = 5.4 \text{ m/s}^2 \leftarrow$



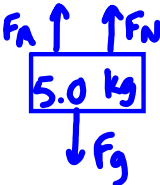
2. a) $F_{\text{net}} = (m_1 + m_2)a = m_2g - m_1g$
 $(1.0 + 0.5)a = 1.0(9.8) - 0.5(9.8)$
 $a = 3.3 \text{ m/s}^2 \curvearrowright$



b) $F_{\text{net}} = m_1g - m_2g = ma \Rightarrow (5.5 + 3.0)a = 5.5(9.8) - 3.0(9.8)$

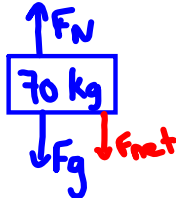


3. a)



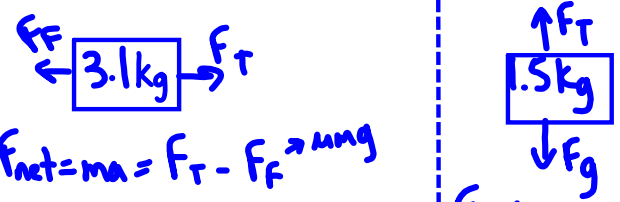
balanced forces: $F_N = F_A + F_g \rightarrow mg$
 $F_{net} = 0 = F_g + F_A - F_N$
 $= 15 + 5.0(9.8)$
 $F_N = 64 \text{ N}$

b)



$F_{net} = F_g - F_N$
 $ma = mg - F_N$
 $70(1) = 70(9.8) - F_N$
 $F_N = 70(9.8) - 70(1)$
 $F_N = 620 \text{ N}$

4. a) $F_{net} = (m_1 + m_2)a = m_2g - F_{f1} \rightarrow \mu F_N = \mu mg$
 $(m + 1.5)2.3 = 1.5(9.8) - 0.14m(9.8)$
 $[2.3 - 0.14(9.8)]m = 1.5(9.8) - 1.5(2.3)$
 $m = 3.1 \text{ kg}$



Left block: $F_{net} = ma = F_T - F_f \rightarrow \mu mg$
 $3.1(2.3) = F_T - 0.14(3.1)(9.8)$

Right block: $F_{net} = ma = F_g - F_T$
 $1.5(2.3) = 1.5(9.8) - F_T$
 $F_T = 11 \text{ N}$

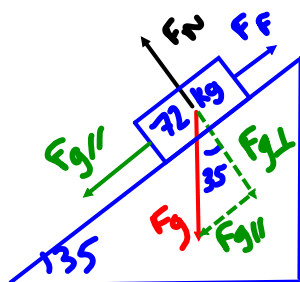
b) $F_{net} = ma = F_g - F_{f2} = m_2g - \mu m_1g$
 $(4.5 + 3.3)a = 3.3(9.8) - 0.17(4.5)(9.8) \Rightarrow a = 3.2 \text{ m/s}^2 \downarrow$



Left block: $ma = F_T - F_f = F_T - \mu mg$
 $4.5(3.2) = F_T - 0.17(4.5)(9.8)$

Right block: $ma = F_g - F_T$
 $3.3(3.2) = 3.3(9.8) - F_T$
 $F_T = 22 \text{ N}$

5.

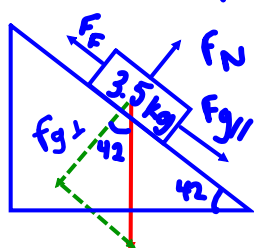


$$F_{net} = ma = F_{g\parallel} - F_f - \mu F_N$$

$$ma = mg \sin \theta - \mu mg \cos \theta$$

$$72a = 72(9.8) \sin 35 - 0.12(72)(9.8) \cos 35$$

$$a = 4.7 \text{ m/s}^2$$

 6. Steady speed $\Rightarrow a = 0 \Rightarrow F_{net} = 0$


$$0 = F_{g\parallel} - F_f$$

$$F_{g\parallel} = F_f$$

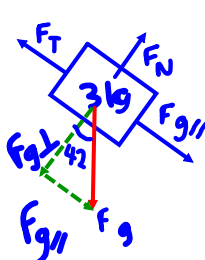
$$a) F_N = mg \cos \theta = 3.5(9.8) \cos 42 = 25 \text{ N}$$

$$b) F_f = F_{g\parallel} = mg \sin \theta = 3.5(9.8) \sin 42 = 23 \text{ N}$$

$$c) F_f = \mu F_N = 23 = \mu 25 \Rightarrow \mu = 0.90$$

$$7a) F_{net} = (m_1 + m_2)a = F_{g\parallel} - F_f \Rightarrow (5+3)a = 3(9.8) \sin 42 - 0.13(5)(9.8)$$

$$a = 1.1 \text{ m/s}^2$$



$$ma = F_T - F_f$$

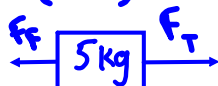
$$5(1.1) = F_T - 0.23(5)(9.8)$$

$$F_T = 17 \text{ N}$$

$$ma = F_{g\parallel} - F_T$$

$$3(1.1) = 3(9.8) \sin 42 - F_T$$

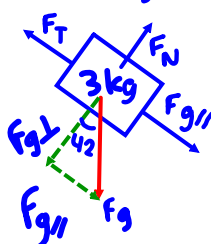
$$b) (3+5)(3.0) = 5(9.8) \sin 42 - \mu(3)(9.8) \Rightarrow \mu = 0.30$$



$$ma = F_T - F_f$$

$$3(3.0) = F_T - 0.30(3)(9.8)$$

$$F_T = 18 \text{ N}$$



$$ma = F_{g\parallel} - F_T$$

$$5(3.0) = 5(9.8) \sin 42 - F_T$$

8. $F_{net} = (m_1 + m_2)a = F_{g1||} - F_{g2||} \Rightarrow (3.47 + 2)a = 3.47(9.8) \sin 30 - 2(9.8) \sin 60$

$a = 0.01 \text{ m/s}^2 \swarrow$

$ma = F_{g||} - F_T$
 $3.47(0.01) = 3.47(9.8) \sin 30 - F_T$

$ma = F_T - F_{g||}$
 $2(0.01) = F_T - 2(9.8) \sin 60$

$F_T = 17 \text{ N}$

9. a) F_{g10} vs. $F_{g30||}$
 $10(9.8)$ vs. $30(9.8) \sin 36$
 98 N vs. 172 N
 *system moves toward 30 kg

$F_{net} = m_{sys} \cdot a = F_{g30} - F_f - F_g$
 $(30 + 10)a = 30(9.8) \sin 36 - 0.22(30)(9.8) \cos 36 - 10(9.8)$

$a = 0.56 \text{ m/s}^2 \swarrow$

$F_{net} = ma = F_{g30} - F_f - F_T$
 $30(0.56) = 30(9.8) \sin 36 - 0.22(30)(9.8) \cos 36 - F_T$

$ma = F_T - F_g$
 $(0.56) = F_T - 10(9.8)$

$F_T = 104 \text{ N}$

b) F_{g20} vs. $F_{g10||} \Rightarrow 20(9.8) \text{ vs. } 10(9.8) \sin 36$
 $196 \text{ N} > 58 \text{ N}$
 *system moves toward 20 kg

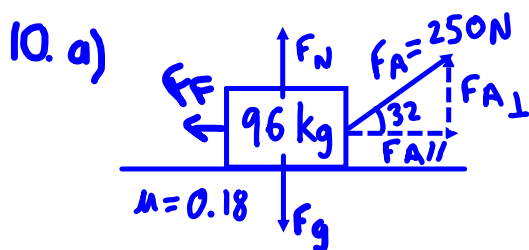
$F_{net} = (m_1 + m_2)a = F_g - F_{g1||} - F_f$
 $(10 + 20)a = 20(9.8) - 10(9.8) \sin 36 - 0.22(10)(9.8) \cos 36$

$a = 5.5 \text{ m/s}^2 \searrow$

$F_{net} = ma = F_T - F_{g1||} - F_f$
 $10(5.5) = F_T - (10)(9.8) \sin 36 - 0.22(10)(9.8) \cos 36$

$F_{net} = ma = F_g - F_T$
 $20(5.5) = 20(9.8) - F_T$

$F_T = 130 \text{ N}$



$$F_{\text{net}} = ma = F_{A||} - F_F \rightarrow \mu F_N$$

$$ma = F_A \cos \theta - \mu (mg - F_A \sin \theta)$$

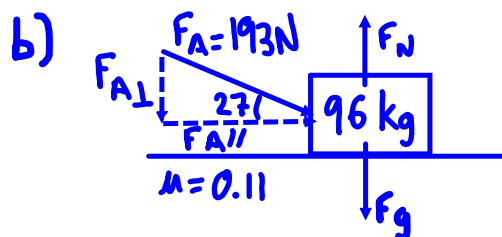
$$96a = 250 \cos 32 - 0.18 [96(9.8) - 250 \sin 32]$$

$$a = 0.69 \text{ m/s}^2 \rightarrow$$

* balanced vertical forces:

$$F_g = F_N + F_{A\perp}$$

$$F_N = F_g - F_{A\perp}$$



$$F_{\text{net}} = ma = F_{||} - F_F$$

$$ma = F \cos \theta - \mu (F_g + F_{\perp})$$

$$68a = 193 \cos 27 - 0.11 [68(9.8) + 193 \sin 27]$$

$$a = 1.3 \text{ m/s}^2$$

$$* F_N = F_g + F_{\perp}$$